

Arjuna JEE (2025)

Maths

DPP: 1

Trigonometric Identities

Q1 Verify

$$\tan^2 30^\circ + \tan^2 45^\circ + \tan^2 60^\circ = 4\frac{1}{3}$$

that:

Q2 Verify that:

$$\sin 30^\circ \cos 60^\circ + \cos 30^\circ \sin 60^\circ = 1.$$

Q3 Verify

that:

$$\cos 45^\circ \cos 60^\circ - \sin 45^\circ \sin 60^\circ = -\frac{\sqrt{3}-1}{2\sqrt{2}}.$$

Q4 $\sin^2 17.5^\circ + \sin^2 72.5^\circ$ is equal to

- (A) $\cos^2 90^\circ$
 (B) $\tan^2 45^\circ$
 (C) $\cos^2 30^\circ$
 (D) $\sin^2 45^\circ$

Q5 Convert $10^\circ 30'$ into radians.

- (A) $\frac{5\pi}{12}$
 (B) $\frac{7\pi}{120}$
 (C) $\frac{11\pi}{12}$
 (D) $\frac{\pi}{12}$

Q6 $45^\circ 15' 30''$ change into degrees.

- (A) $\left(\frac{5431}{120}\right)^\circ$
 (B) $\left(\frac{5400}{120}\right)^\circ$
 (C) 45°
 (D) $\left(\frac{543}{120}\right)^\circ$

Q7 If the arcs of the same length in two circles subtend angles 65° and 110° at the centre, then find the ratio of their radii.

- (A) $\frac{22}{13}$
 (B) $\frac{11}{13}$
 (C) $\frac{5}{13}$

(D) $\frac{110}{13}$

Q8 Find the length of an arc of a circle of radius 5 cm subtending a central angle measuring 15° .

- (A) $\frac{11\pi}{12}$
 (B) $\frac{7\pi}{12}$
 (C) $\frac{\pi}{12}$
 (D) $\frac{5\pi}{12}$

Q9 The difference between the two acute angles of a right-angled triangle is $\frac{\pi}{10}$ radians. Express the angles in degree.

- (A) $63^\circ, 27^\circ$
 (B) $54^\circ, 36^\circ$
 (C) $33^\circ, 57^\circ$
 (D) $75^\circ, 25^\circ$

Q10 Express $45^\circ 20' 10''$ in radians.

- (A) 0.81 radian
 (B) 0.336 radian
 (C) 0.79 radian
 (D) 0.10 radian

Q11 The angle subtended at the centre of circle of radius 3 meters by an arc of length 1 meter is equal to

- (A) 20°
 (B) 60°
 (C) $\frac{1}{3}$ radian
 (D) 3 radian

Q12 If $\tan \theta = \frac{-4}{3}$, then, $\sin \theta$ is equal to

- (A) $\frac{-4}{5}$ but not $\frac{4}{5}$



- (B) $\frac{-4}{5}$ or $\frac{4}{5}$
 (C) $\frac{4}{5}$ but not $\frac{-4}{5}$
 (D) None of these

Q13 Find quadrant in which following angles lie

- (i) 765°
 (ii) 5253°
 (iii) -8054°
 (iv) 2030°
 (v) -7156°

Q14 If $\sin \theta = \frac{24}{25}$ and θ lies in the second quadrant, then $\sec \theta + \tan \theta =$

- (A) -3 (B) -5
 (C) -7 (D) -9

Q15 Simplify the following:

$$\frac{\sin(180^\circ + \theta) \cos(360^\circ - \theta) \tan(270^\circ - \theta)}{\sec^2(90^\circ + \theta) \tan(-\theta) \sin(270^\circ + \theta)}$$

Q16 Simplify each of the following:

$$\frac{\tan(90^\circ - \theta) \sec(180^\circ - \theta) \sin(-\theta)}{\sin(180^\circ + \theta) \cot(360^\circ - \theta) \operatorname{cosec}(90^\circ - \theta)}$$

Q17 Simplify each of the following:

$$\frac{\cos(2\pi + \theta) \operatorname{cosec}(2\pi - \theta) \tan(\pi/2 + \theta)}{\sec(\pi/2 + \theta) \cos \theta \cot(\pi + \theta)}$$

Q18 Simplify each of the following:

$$\frac{\operatorname{cosec}(90^\circ + \theta) + \cot(450^\circ + \theta)}{\operatorname{cosec}(450^\circ - \theta) - \tan(180^\circ + \theta)} + \frac{\tan(180^\circ + \theta) + \sec(180^\circ - \theta)}{\tan(360^\circ + \theta) - \sec(-\theta)}$$

Q19 If A lies in the second quadrant and $3 \tan A + 4 = 0$, then the value of $2 \cot A - 5 \cos A + \sin A$ is equal to

- (A) $\frac{-53}{10}$
 (B) $\frac{23}{10}$
 (C) $\frac{37}{10}$
 (D) $\frac{7}{10}$

Q20 If $\tan \theta + \sec \theta = \frac{5}{3}$ and ' θ ' is acute angle then value of $\sin \theta$ is $\frac{8}{p}$, then the value of p is

- (A) 12 (B) 71
 (C) 17 (D) 20

Q21 $3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 \cos^6 x$
 $+ 4(\sin^6 x + \cos^6 x)$
 is equal to

- (A) 11 (B) 12
 (C) 13 (D) 14



Answer Key

- Q1 hence proved
Q2 hence proved
Q3 Hence proved.
Q4 (B)
Q5 (B)
Q6 (A)
Q7 (A)
Q8 (D)
Q9 (B)
Q10 (C)
Q11 (C)
Q12 (B)

- Q13 i) First quadrant
 ii) Third quadrant
 iii) Third quadrant
 iv) Third quadrant
 v) First quadrant
Q14 (C)
Q15 $-\sin(\theta)(\cos(\theta))^2$
Q16 1
Q17 -1
Q18 2
Q19 (B)
Q20 (C)
Q21 (C)



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